LGL1: Pathology of oral cavity and salivary glands

By:

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Terminology: The following terms are used to describe localized inflammation of oral cavity:

- Cheilitis (lips)
- Gingivitis (gum)
- Glossitis (tongue)
- Stomatitis (oral mucosa)

Oral cavity

Infections of Oral Cavity:

Causes:

- **Viral:** HSV, EBV, HPV
- **Bacterial:** Ludwig angina, diphtheria, syphilis
- **Fungal:** Candidiasis.

Viral infections:

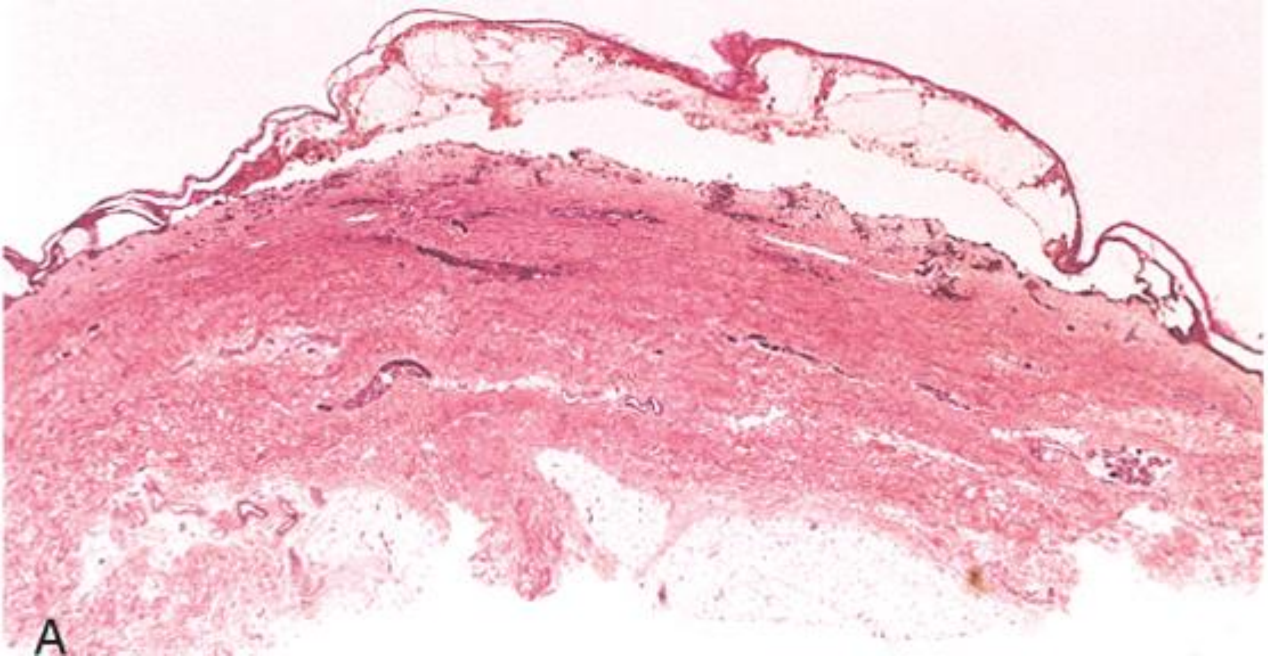
Herpes Simplex Virus (HSV) Infections:

- Caused by (HSV-1), with an increase in those caused by HSV-2 (genital herpes).
- Primary infections occur in children & are often asymptomatic.

- Recurrent herpetic stomatitis:
- Occurs in adults who harbor latent HSV-1, and the virus can be reactivated, resulting in **"Cold sore".**
- It appears as groups of small (1-3 mm) vesicles on lips, nasal orifices, buccal mucosa, gingiva & hard palate.
- It resolves within 7-10 days, but may persist in immuno-compromised patients.

- Factors associated with HSV reactivation include:-
- Trauma.
- Allergies.
- Exposure to ultraviolet light.
- Upper respiratory tract infections.
- Pregnancy & menstruation.
- Immunosuppression.
- Exposure to extremes of temperature.

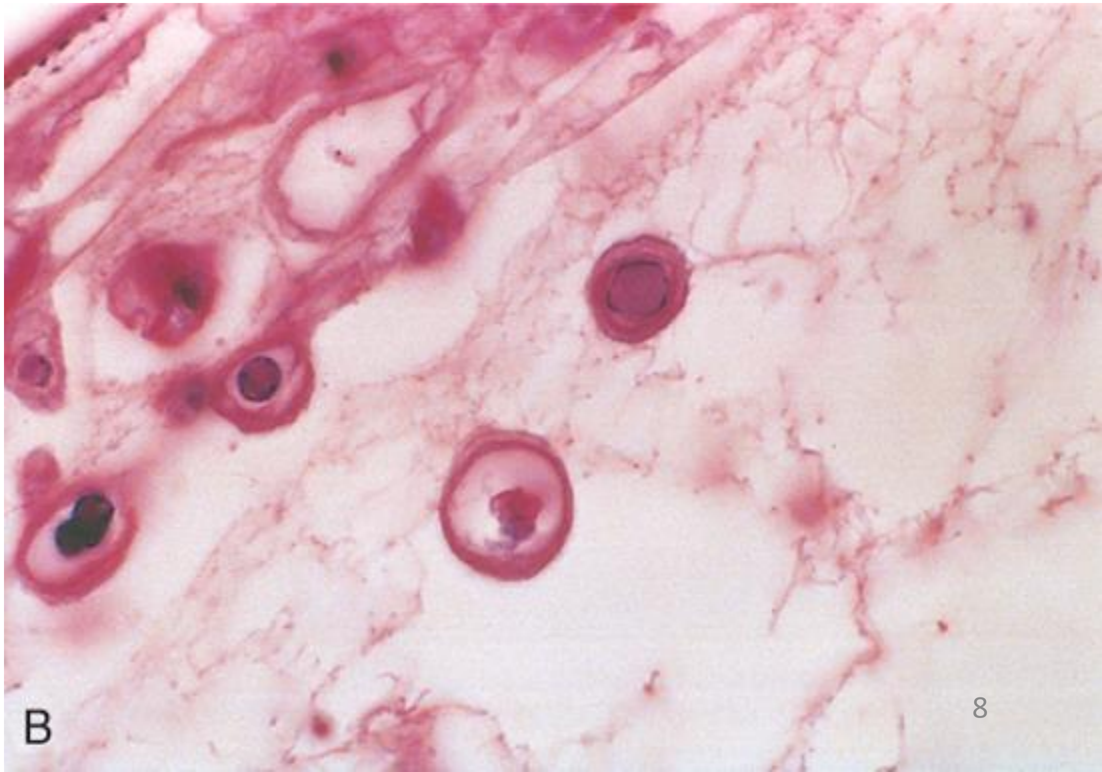
orofacial herpetic infections



<u>**Microscope:**</u>

A: Herpesvirus blister in mucosa.

B: cells from blister in A, showing glassy intranuclear HSV inclusion bodies.



Inflammatory/ Reactive Lesions

Aphthous Ulcers (Canker Sores):

- Common superficial mucosal ulcerations.
- Affect up to 40% of population.
- Extremely painful and recurrent superficial oral mucosal ulcerations of unknown etiology.
- It tend to be prevalent within certain families.
- May associated with immunologic disorders as celiac disease, inflammatory bowel disease & behçet disease.

- Predisposing factors:
  - Cessation of smoking.
  - Stress
  - Fever
  - Trauma
  - Endocrine factors in some women: increase in luteal phase of menstrual cycle, and may regress in pregnancy
  - Allergies to food.

- Presentation:
  - They appear as single or multiple ulcers covered by a thin fibrinopurulent exudate surrounded by an erythematatous halo.
  - Typically resolve spontaneously in 7 to 10 days, but may persist for weeks.

Proliferative and Neoplastic lesions:

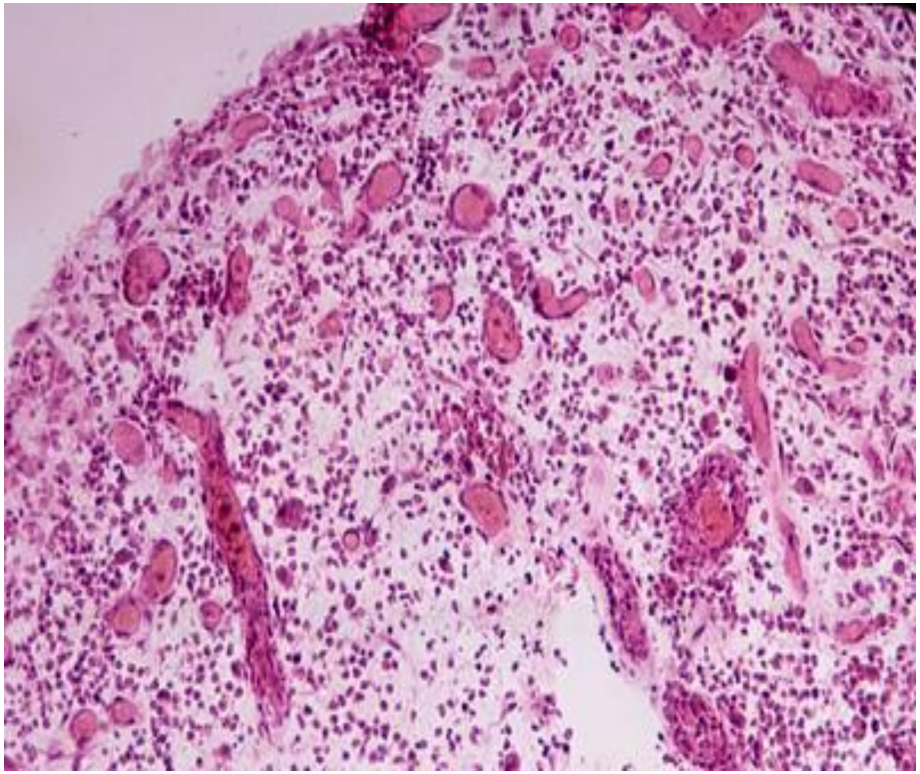
Fibrous Proliferative Lesions:

Fibromas:

- Are submucosal nodular fibrous tissue masses that are formed when chronic irritation results in reactive connective tissue hyperplasia.
- They occur most often on the buccal mucosa along the bite line.

Pyogenic Granuloma:

- It is typically found on the gingiva of children, young adults, and pregnant women (pregnancy tumor).
- The etiology is unknown, possibly a response of tissues to minor trauma or chronic irritation.
- These lesions are richly vascular and typically are ulcerated,
- It may regress spontaneously.



Grossly: it appear as polypoidal lesion that usually ulcerate

Histologically: it composed of highly vascular proliferation of organizing granulation tissue.

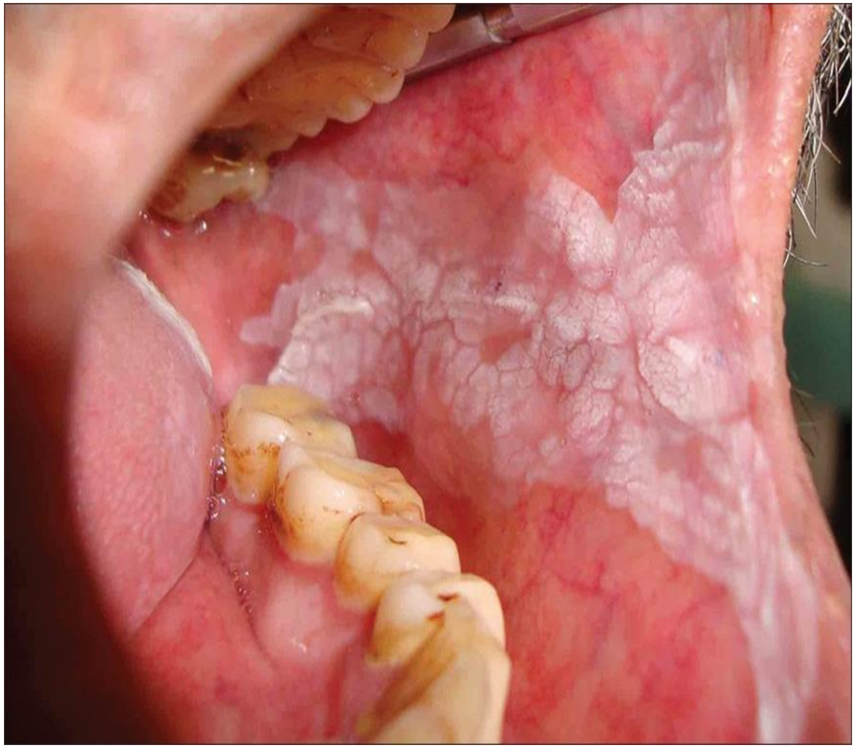
Leukoplakia and Erythroplakia:

- May be seen in adults typically between the ages of 40 & 70 years with a 2 : 1 male preponderance.
- Although the etiology is multifactorial, tobacco use (cigarettes, pipes, cigars, and chewing tobacco) is the most common risk factor.

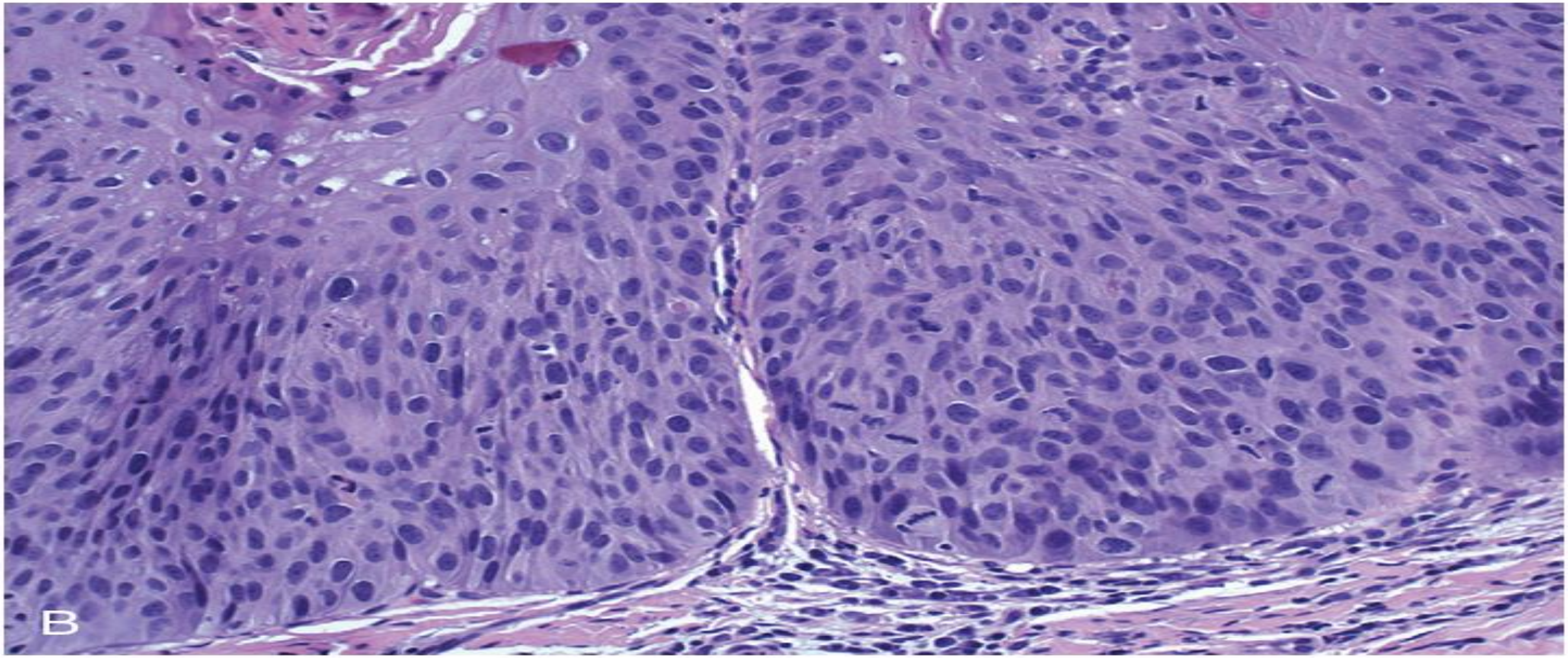
Leukoplakia:

- A white patch or plaque that cannot be scraped off & cannot be characterized clinically or pathologically as any other disease.
- The etiology is unknown.
- About 3% of world's population have leukoplakia.
- White patches caused by obvious irritation or diseases as lichen planus & candidiasis are not leukoplakias.

- 5-25% are premalignant and may progress to squamous cell carcinoma.
- Thus, until proved otherwise by means of histologic evaluation, all leukoplakias must be considered precancerous.



- Histopathology: leukoplakia may show a spectrum of changes, from increased surface keratinization without dysplasia to invasive keratinizing SCC.



Histologic appearance of leukoplakia showing dysplasia, characterized by nuclear and cellular pleomorphism and loss of normal maturation.

Erythroplakia:

- Is a red, velvety, possibly eroded area that is flat or slightly depressed relative to the surrounding mucosa.
- It is associated with a much greater risk of malignant transformation than leukoplakia.

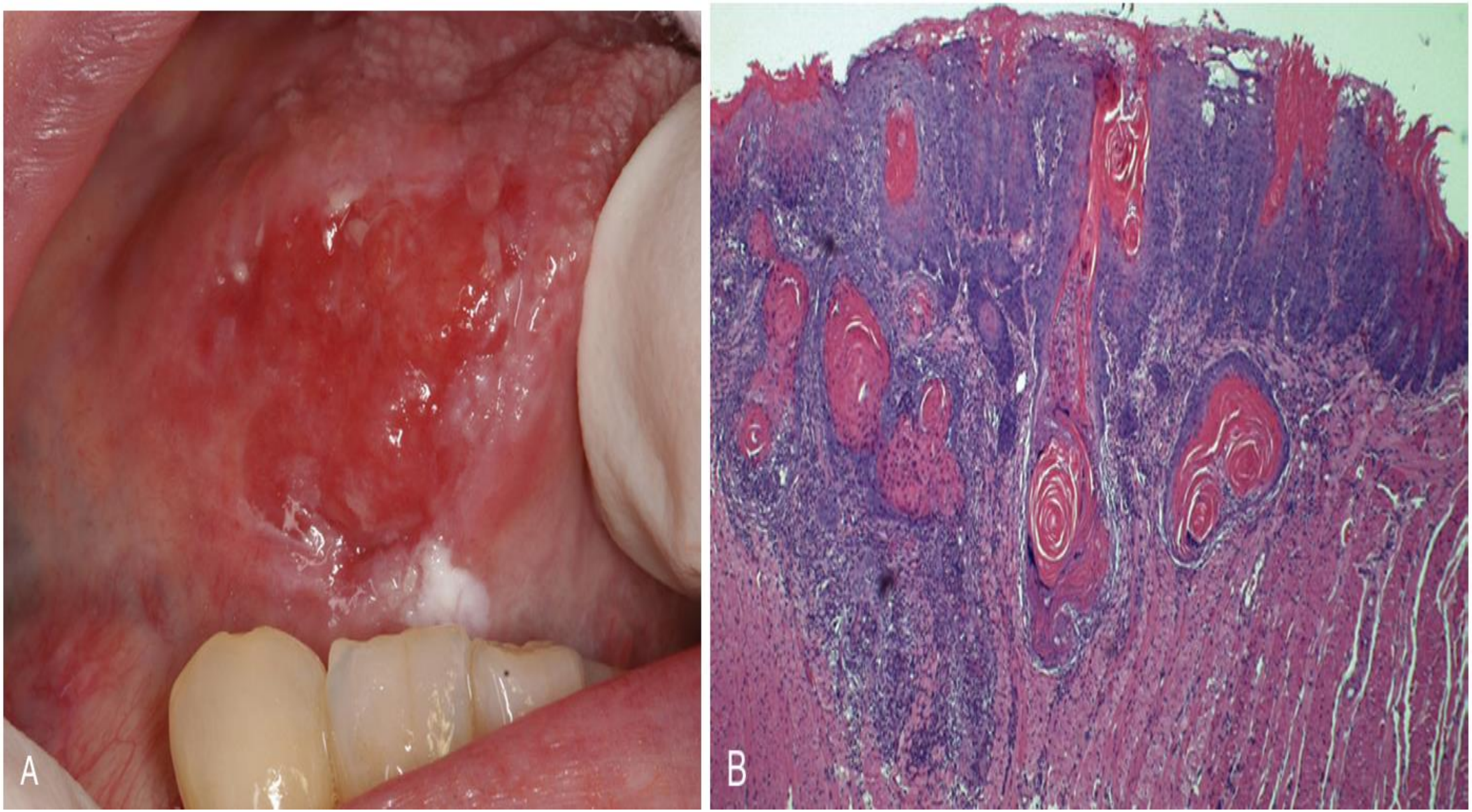
Squamous Cell Carcinoma (SCC):

- Account for about 95% of oral cavity cancers.
- It is aggressive tumor with poor survival.
- The cervical LN are the most common sites of regional metastasis; frequent sites of distant metastases include the mediastinal LN, lungs, and liver.

- Etiology and Risk factors:

Factor	Risk of transformation
Leukoplakia	3% to 25%
Erythroplakia	More than 50%
Tobacco use	Best-established influence, particularly pipe smoking and smokeless tobacco
Human papillomavirus types 16 and 18	Identified in 30% to 50% of cases; probably have a role in a subset of cases
Alcohol abuse	Weaker influence than tobacco use, but the two habits interact to greatly increase risk
Protracted irritation	Weakly associated

Morphology:



Oral squamous cell carcinoma: A, Clinical appearance demonstrating ulceration and induration of the oral mucosa. B, Histologic appearance demonstrating numerous nests and islands of malignant keratinocytes invading the underlying connective tissue stroma.

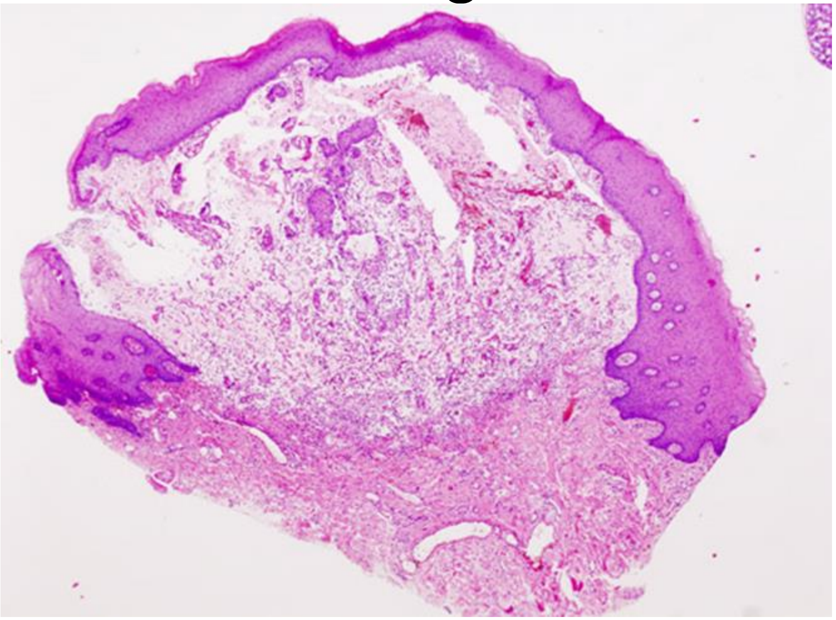
Diseases Of Salivary Glands

Sialadenitis:

- Is inflammation of the SG, may be induced by:
  - Trauma
  - Viral infection e.g. mumps
  - Bacterial infection: by staph or strept in case of duct obstruction by stone or food particles or due to dehydration.
  - Autoimmune disease e.g. Sjögren syndrome.

Mucocele

- Is the most common lesion of the minor SG.
- Is a cystic lesion filled with mucous.
- It results from:
  - Either blockage of a SG duct (Retention mucocele)
  - Or rupture of a SG duct (Extravasation mucocele) with consequent leakage of saliva into surrounding CT stroma.



Neoplasms of Salivary Glands

- Uncommon: 2% of head and neck neoplasms
- Distribution:
  - Parotid: 80% overall; 80% benign
  - Submandibular: 15% overall; 50% benign
  - Sublingual/Minor SG: 5% overall; 40% benign
- The smaller the SG involved, the more likely the tumor in it will be malignant.

Table 14–1 Histopathologic Classification and Prevalence of the Most Common Benign and Malignant Salivary Gland Tumors

Benign	Malignant
Pleomorphic adenoma (50%)	Mucoepidermoid carcinoma (15%)
Warthin tumor (5%)	Acinic cell carcinoma (6%)
Oncocytoma (2%)	Adenocarcinoma NOS (6%)
Cystadenoma (2%)	Adenoid cystic carcinoma (4%)
Basal cell adenoma (2%)	Malignant mixed tumor (3%)

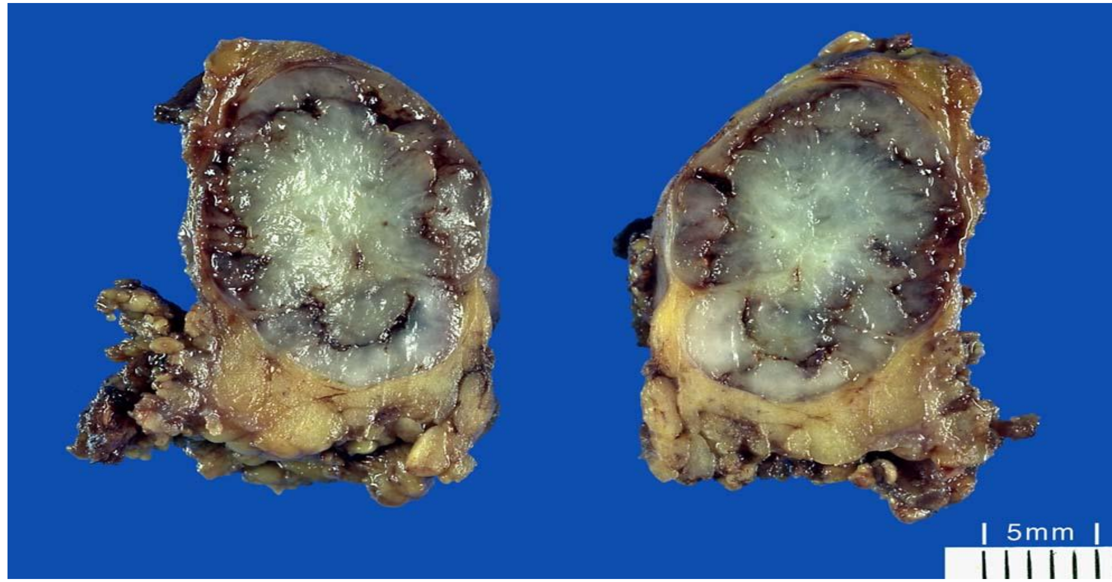
NOS, not otherwise specified.

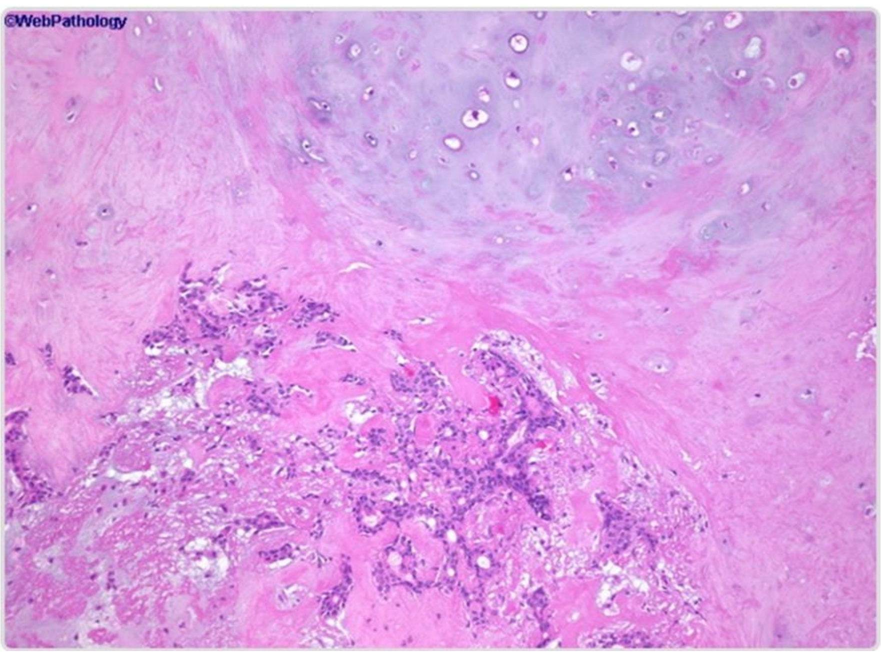
Data from Ellis GL, Auclair PL, Gnepp DR: Surgical Pathology of the Salivary Glands, Vol 25: Major Problems in Pathology, Philadelphia, WB Saunders, 1991.

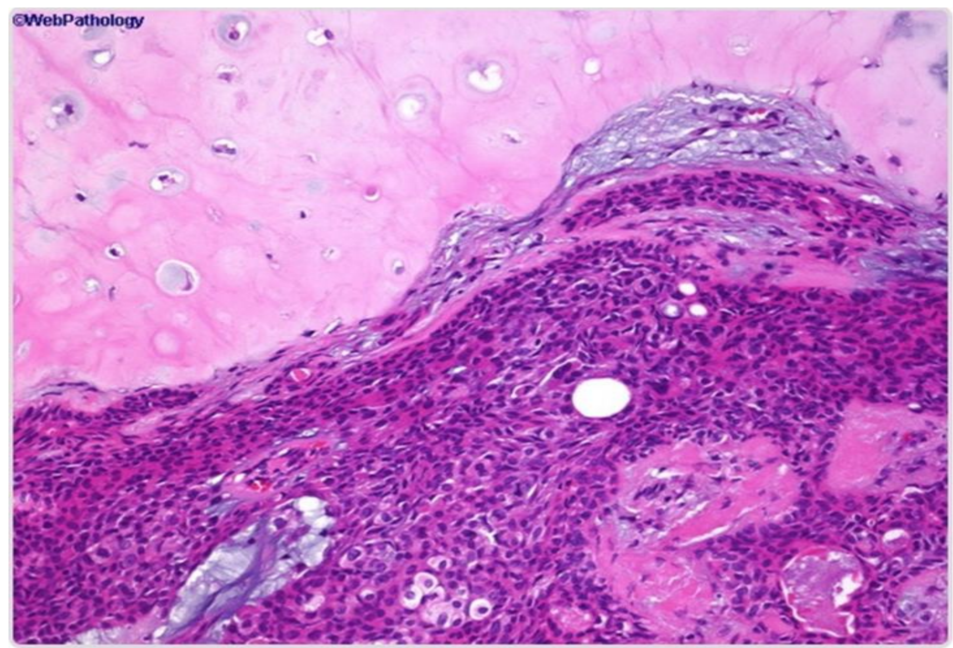
Pleomorphic Adenoma (PA):

- It is more common in middle aged females.
- Present as painless, slow-growing, rounded, well demarcated masses.
- Also called Benign mixed tumor of SG.
- It is common (about 60% of tumors in parotid gland).
- Consist of a mixture of ductal (epithelial) and myoepithelial cells; So they exhibit both epithelial and mesenchymal differentiation.

- Epithelial elements are dispersed throughout the matrix.
- Mesenchymal elements which may contain variable mixtures of myxoid, hyaline, chondroid (cartilaginous), and even osseous tissue.
- PA recur if incompletely excised.
- Carcinoma arising in PA is referred to carcinoma ex pleomorphic adenoma or malignant mixed tumor.







Pleomorphic Adenoma (Benign mixed tumor)

Mucoepidermoid Carcinoma (MEC):

- The most common form of primary malignant tumor of the SGs.
- Composed of variable mixtures of squamous cells, mucus-secreting cells, and intermediate cells.